

Chapter 5

Basic SQL 1

1

Software

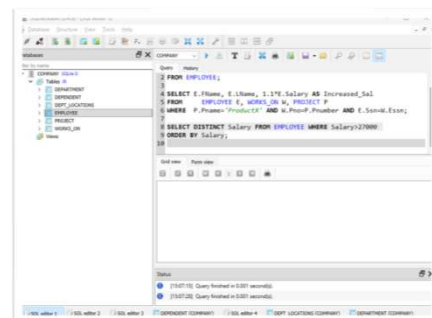


SQLite 3.47.0 (command line)
<https://www.sqlite.com>



SQLiteStudio 3.4.9 (IDE)
<https://sqlitestudio.pl>

```
SQLite version 3.47.0 2020-08-01 16:38:22
Enter "help" for usage hints.
SQLite > .open test.db
SQLite > .help
SQLite > .mode text
SQLite > .tables
table
SQLite > .dump table
CREATE TABLE table (
  id integer PRIMARY KEY,
  name text
);
INSERT INTO table VALUES (1, 'John');
INSERT INTO table VALUES (2, 'Jane');
INSERT INTO table VALUES (3, 'Bob');
INSERT INTO table VALUES (4, 'Alice');
INSERT INTO table VALUES (5, 'Charlie');
```



2

Software (for the adventurous)



Oracle MySQL Workbench 8.0.40
<https://www.mysql.com>



Oracle SQL Developer 24.3
<https://www.oracle.com/database/sqldeveloper>



Microsoft SQL Server 2022
<https://www.microsoft.com/sql-server>

3

The CREATE TABLE Command

Creates a new relation

- Specify name of table
- Specify attributes, their types and initial constraints

```
CREATE TABLE EMPLOYEE (  
  Fname      VARCHAR(15)      NOT NULL,  
  Minit      CHAR,  
  Lname      VARCHAR(15)      NOT NULL,  
  Ssn        CHAR(9)          NOT NULL,  
  Bdate      DATE,  
  Address    VARCHAR(30),  
  Sex        CHAR,  
  Salary     DECIMAL(10,2),  
  Super_ssn  CHAR(9),  
  Dno        INT              NOT NULL,  
  
  PRIMARY KEY (Ssn),  
  
  CONSTRAINT EMPSUPERFK  
    FOREIGN KEY (Super_ssn) REFERENCES EMPLOYEE(Ssn)  
    ON DELETE SET NULL  
    ON UPDATE CASCADE,  
  
  CONSTRAINT EMPDEPTFK  
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dnumber)  
    ON DELETE SET DEFAULT  
    ON UPDATE CASCADE  
);
```

4

Attribute Data Types and Domains

Numeric data types

- Integer numbers:
`INTEGER`
`INT`
`SMALLINT`
- Floating-point (real) numbers:
`FLOAT` or `REAL`
`DOUBLE PRECISION`

5

Attribute Data Types

Character string data types

- Fixed length:
`CHAR (n)`
`CHARACTER (n)`
- Varying length:
`VARCHAR (n)`
`CHAR VARYING (n)`
`CHARACTER VARYING (n)`

6

Attribute Data Types

Bit-string data types

- Fixed length: **BIT (n)**
- Varying length: **BIT VARYING (n)**

Boolean data type

- Values of **TRUE** or **FALSE** or **NULL**

7

Attribute Data Types

DATE data type

- Components are **YEAR**, **MONTH**, and **DAY** in the form YY-MM-DD
- Multiple mapping functions available in RDBMSs to change date formats

TIME data type

- Components are **HOURS**, **MINUTES**, and **SECONDS** in the form HH:MM:SS

8

Domains and Types

DOMAIN

- Name used with the attribute specification
- Makes it easier to change the data type for a domain that is used by numerous attributes
- Improves schema readability

```
CREATE DOMAIN SSN_TYPE AS CHAR(9);
```

TYPE

```
CREATE TYPE USA_ADDR_TYPE AS (  
    STREET_ADDR ROW(  
        NUMBER          VARCHAR(5),  
        STREET_NAME     VARCHAR(25),  
        APT_NO          VARCHAR(5),  
        SUITE_NO         VARCHAR(5) ),  
    CITY                VARCHAR(25),  
    STATE               CHAR(2),  
    ZIP                CHAR(5)  
);
```

9

Attribute Constraints

Restrictions on attribute domains:

- Default value of an attribute
 - **DEFAULT <value>**
- **NULL** is not permitted for a particular attribute (**NOT NULL**)

CHECK clause

```
Dnumber INT NOT NULL  
CHECK (Dnumber > 0 AND Dnumber < 21);
```

```
ZIP INT NOT NULL  
CHECK (ZIP > 0 AND ZIP < 100000)
```

10

Keys

PRIMARY KEY clause specifies primary key:

In attribute definition:

```
Dnumber INT PRIMARY KEY;
```

Or separately:

```
PRIMARY KEY (Dnumber) ,
```

UNIQUE clause specifies secondary keys:

In attribute definition:

```
Dname VARCHAR(15) UNIQUE;
```

Or separately:

```
UNIQUE (Dname) ,
```

11

Key-related Constraints

Basic constraints:

The relational model has three basic constraint types that are supported in SQL:

- **Key** constraint: A primary key value must be unique
- **Entity Integrity** constraint: A primary key value cannot be null
- **Referential integrity** constraints: The “foreign key” must have a value that is already present as a primary key, or may be null.

12

Foreign Keys

FOREIGN KEY clause

Default: reject update on violation

```
FOREIGN KEY (Super_ssn) REFERENCES EMPLOYEE (Ssn)
```

```
FOREIGN KEY (Dno) REFERENCES DEPARTMENT (Dnumber)
```

13

Violation of Referential Integrity Constraints

FOREIGN KEY clause

Attach **referential triggered action** clause

Options include:

```
SET NULL  
CASCADE  
SET DEFAULT
```

```
FOREIGN KEY (Super_ssn) REFERENCES  
EMPLOYEE (Ssn)  
ON DELETE SET NULL  
ON UPDATE CASCADE
```

Action taken by the DBMS for

SET NULL or **SET DEFAULT** is the same for both **ON DELETE** and **ON UPDATE**

CASCADE option suitable for “relationship” relations

14

Naming Constraints

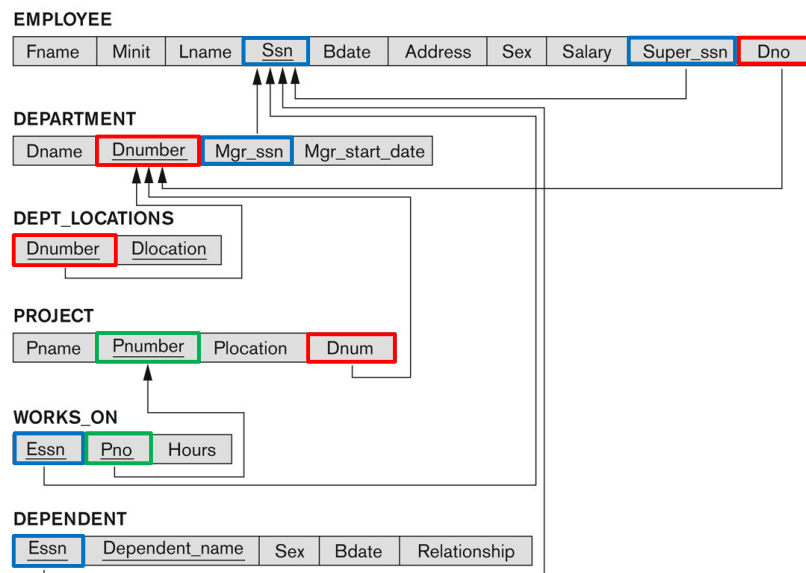
Use the keyword **CONSTRAINT**

- Name a constraint
- Useful for later altering

```
CONSTRAINT EMPSUPERFK  
FOREIGN KEY (Super_ssn) REFERENCES EMPLOYEE (Ssn)  
ON DELETE SET NULL  
ON UPDATE CASCADE
```

15

Referential Integrity Constraints for COMPANY



16

Populated Database State for COMPANY

EMPLOYEE										DEPT_LOCATIONS		WORKS_ON		
Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno	Dnumber	Dlocation	Essn	Pno	Hours
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5	1	Houston	123456789	1	32.5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5	4	Stafford	123456789	2	7.5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4	5	Bellaire	666884444	3	40.0
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4	5	Sugarland	453453453	1	20.0
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5	5	Houston	453453453	2	20.0
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5			333445555	2	10.0
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4			333445555	3	10.0
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1			333445555	10	10.0

DEPARTMENT			
Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPENDENT				
Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

PROJECT			
Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

17



`.read CREATE_COMPANY.sql`

`.read IMPORT_COMPANY_DATA.sqlite`

<code>.import EMPLOYEE.csv</code>	EMPLOYEE	-csv
<code>.import DEPARTMENT.csv</code>	DEPARTMENT	-csv
<code>.import DEPT_LOCATIONS.csv</code>	DEPT_LOCATIONS	-csv
<code>.import DEPENDENT.csv</code>	DEPENDENT	-csv
<code>.import PROJECT.csv</code>	PROJECT	-csv
<code>.import WORKS_ON.csv</code>	WORKS_ON	-csv

`.save COMPANY.db`



add a database

```

CREATE TABLE EMPLOYEE
(
    Fname          VARCHAR(15) NOT NULL,
    Minit          CHAR,
    Lname          VARCHAR(15) NOT NULL,
    Ssn            CHAR(9) NOT NULL,
    Bdate          DATE,
    Address         VARCHAR(30),
    Sex            CHAR,
    Salary         DECIMAL(10,2),
    Super_ssn      CHAR(9),
    Dno            INT NOT NULL,

    PRIMARY KEY (Ssn),
    CONSTRAINT EMPSUPERFK
    FOREIGN KEY (Super_ssn) REFERENCES EMPLOYEE(Ssn)
        ON DELETE SET NULL
        ON UPDATE CASCADE,
    CONSTRAINT EMPDEPTFK
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dnumber)
        ON DELETE SET DEFAULT
        ON UPDATE CASCADE
);

CREATE TABLE DEPARTMENT (
    Dname          VARCHAR(15) NOT NULL,
    Dnumber        INT NOT NULL,
    Mgr_ssn        CHAR(9) NOT NULL,
    Mgr_start_date DATE,

    PRIMARY KEY (Dnumber),
    UNIQUE (Dname),
    FOREIGN KEY (Mgr_ssn) REFERENCES EMPLOYEE(Ssn)
);

CREATE TABLE DEPT_LOCATIONS (
    Dnumber        INT NOT NULL,
    Dlocation      VARCHAR(15) NOT NULL,

    PRIMARY KEY (Dnumber,Dlocation),
    FOREIGN KEY (Dnumber) REFERENCES DEPARTMENT(Dnumber)
);

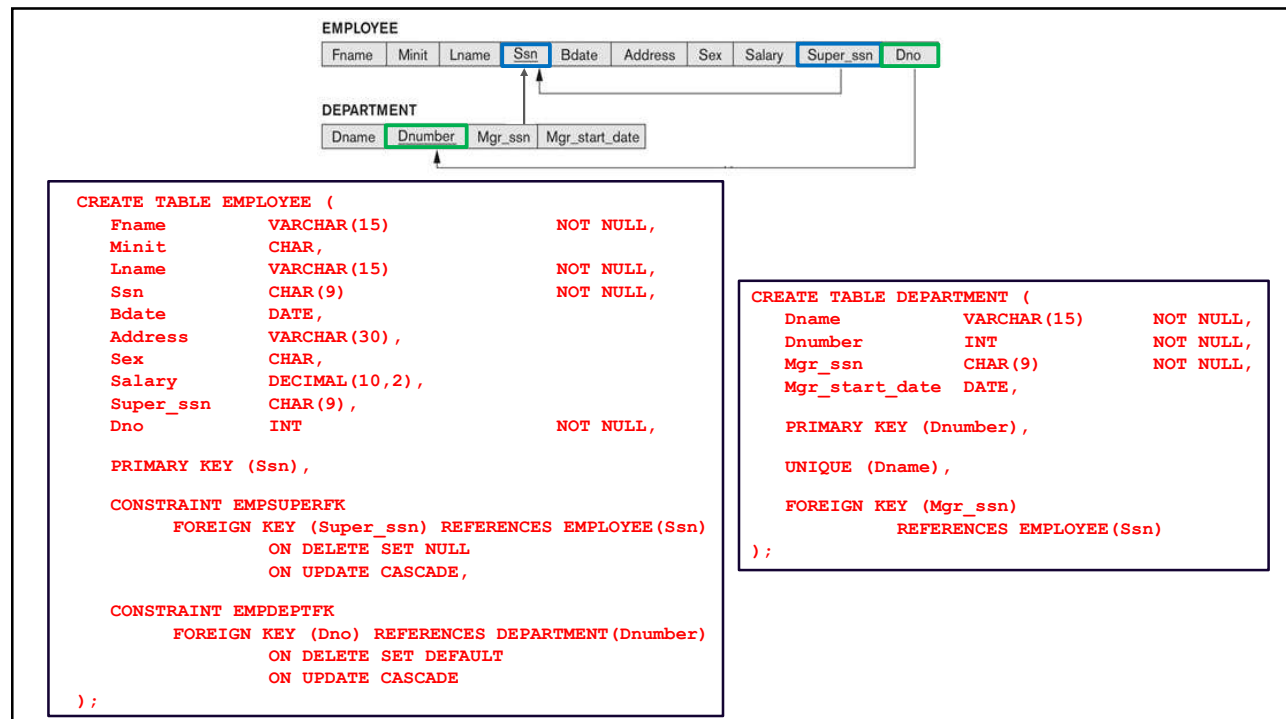
```

18

Name	Data type	Primary Key	Foreign Key	Unique	Check	Not NULL	Collate	Generated	Default value
2 Minit	CHAR								NULL
3 Lname	VARCHAR (15)								NULL
4 Ssn	CHAR (9)								NULL
5 Bdate	DATE								NULL
6 Address	VARCHAR (30)								NULL
7 Sex	CHAR								NULL
8 Salary	DECIMAL (10, 2)								NULL

Type	Name	Details
1 PRIMARY KEY	(Ssn)	
2 FOREIGN KEY EMPSUPERFK	(Super_ssn)	REFERENCES EMPLOYEE (Ssn) ON DELETE SET NULL ON UPDATE CASCADE
3 FOREIGN KEY EMPDEPTFK	(Dno)	REFERENCES DEPARTMENT (Dnumber) ON DELETE SET DEFAULT ON UPDATE ...

19



20

```
CREATE TABLE DEPT_LOCATIONS (
    Dnumber    INT            NOT NULL,
    Dlocation  VARCHAR(15)   NOT NULL,

    PRIMARY KEY (Dnumber,Dlocation),

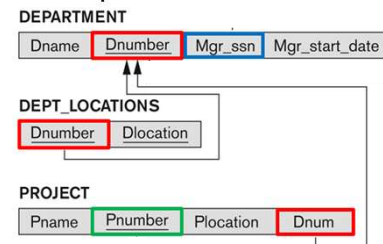
    FOREIGN KEY (Dnumber) REFERENCES DEPARTMENT(Dnumber)
);
```

```
CREATE TABLE PROJECT (
    Pname      VARCHAR(15)   NOT NULL,
    Pnumber    INT            NOT NULL,
    Plocation  VARCHAR(15),
    Dnum       INT            NOT NULL,

    PRIMARY KEY (Pnumber),

    UNIQUE (Pname),

    FOREIGN KEY (Dnum) REFERENCES DEPARTMENT(Dnumber)
);
```



21

```
CREATE TABLE WORKS_ON (
    Essn       CHAR(9)        NOT NULL,
    Pno        INT            NOT NULL,
    Hours      DECIMAL(3,1)   NOT NULL,

    PRIMARY KEY (Essn,Pno),

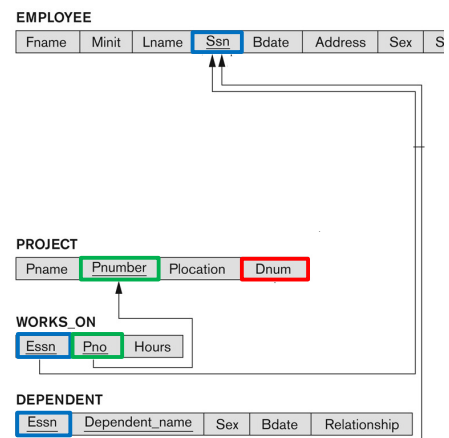
    FOREIGN KEY (Essn) REFERENCES EMPLOYEE(Ssn),

    FOREIGN KEY (Pno) REFERENCES PROJECT(Pnumber)
);
```

```
CREATE TABLE DEPENDENT (
    Essn       CHAR(9)        NOT NULL,
    Dependent_name VARCHAR(15) NOT NULL,
    Sex        CHAR,
    Bdate      DATE,
    Relationship VARCHAR(8),

    PRIMARY KEY (Essn,Dependent_name),

    FOREIGN KEY (Essn) REFERENCES EMPLOYEE(Ssn)
);
```



22

INSERT

Specify the relation name and a list of values (including nulls) for the tuple.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
-------	-------	-------	-----	-------	---------	-----	--------	-----------	-----

INSERT INTO EMPLOYEE VALUES ('John','B','Smith','123456789','1965-01-09','731 Fondren, Houston TX','M',30000,'333445555',5);

23

INSERT

INSERT INTO EMPLOYEE VALUES ('Richard','K','Marini','333445555','1962-12-30','98 Oak Forest, Katy,TX','M',37000,'987654321',4);

INSERT INTO DEPARTMENT VALUES ('Research',5,'333445555','1988-05-22');

INSERT INTO PROJECT VALUES ('ProductX',1,'Bellaire',5);

INSERT INTO WORKS_ON VALUES ('123456789',1,32.5);

INSERT INTO DEPENDENT VALUES ('333445555','Alice','F','1986-04-0','Daughter');

INSERT INTO DEPT_LOCATIONS VALUES (1,'Houston');

24

SELECT

SELECT is the basic statement for retrieving information from a database

```
SELECT SSN
FROM EMPLOYEE;
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1



Ssn
123456789
333445555
999887777
987654321
666884444
453453453
987987987
888665555

25

SELECT-FROM-WHERE

```
SELECT <attribute_list>
FROM <table_list>
WHERE <condition>;
```

<attribute_list> - list of attributes whose values are to be retrieved.

<table_list> - list of the relation names from which the attributes are retrieved.

<condition> - conditional expression identifying the tuples to be considered.

26

Example: SELECT from single relation

Query: Retrieve the birth date and address of the employee(s) whose name is 'John B. Smith'.

```
SELECT Bdate, Address
FROM EMPLOYEE
WHERE Fname='John' AND Minit='B' AND Lname='Smith';
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Bdate	Address
1965-01-09	731 Fondren, Houston, TX

27

Example: SELECT by “joining” two relations

Query: Retrieve the name and address of all employees who work for the 'Research' department.

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

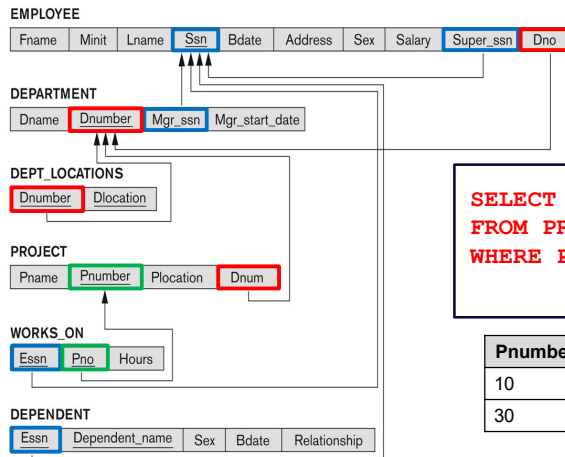
```
SELECT Fname, Minit, Lname, Address
FROM EMPLOYEE, DEPARTMENT
WHERE Dname='Research' AND Dnumber=Dno;
```

Fname	Minit	Lname	Address
John	B	Smith	731 Fondren, Houston, TX
Franklin	T	Wong	638 Voss, Houston, TX
Ramesh	K	Narayan	975 Fire Oak, Humble, TX
Joyce	A	English	5631 Rice, Houston, TX

28

Example: SELECT by “joining” three relations

Query: For every project located in ‘Stafford’, list the project number, the controlling department number, and the department manager’s last name, address, and birth date.



```
SELECT Pnumber, Dnum, Lname, Address, Bdate
FROM PROJECT, EMPLOYEE, DEPARTMENT
WHERE Plocation='Stafford' AND Dnum=Dnumber
AND Mgr_ssn=Ssn
```

Pnumber	Dnum	Lname	Address	Bdate
10	4	Wallace	291 Berry, Bellaire, TX	1941-06-20
30	4	Wallace	291 Berry, Bellaire, TX	1941-06-20